An empirical analysis of the Buy Box assignment in the Amazon platform

Abstract

Amazon is the leading Western company in online commerce, being at the same time retailer, supplier, seller, and owner of the largest e-commerce trading platform. In this work, we perform an empirical analysis to assess which features are the most relevant in the Amazon platform for the Buy Box assignment, i.e., the first and predefined selling option proposed to the customer during the online shopping process of a product. Thanks to a newly developed data extraction and machine learning analysis system, we collect and process data from different products offered in the platform and during three distinct time periods. The obtained results indicate that the most important features are the ones related to the customers' ratings, to the use of Amazon's logistic services (i.e., fba), and in Amazon being the seller. Finally, we release our system publicly for allowing researchers to perform further studies and analyses.

Keywords: digital markets, e-commerce, Amazon, reverse engineering of algorithms, machine learning

1 Introduction

As e-commerce becomes increasingly relevant in contemporary economies, focusing on the analysis of Amazon, the leading Western platform in online commerce, is a significant research activity. Amazon represents one of the big tech players: indeed, it is vertically integrated into more than one sector and its business models are complex. Most importantly, Amazon is at the same time a digital marketplace essential for many economic actors, where it plays the role of supplier, retailer, seller, and owner of the platform.

The aim of this work is to understand which features are the most relevant for the Amazon platform when assigning to a third party seller the Buy Box, i.e., the first and predefined selling option proposed to the customer during the online shopping process of a product. It is the box that appears on the upper position of the page of a specific product and through which the majority of consumers conclude their purchase. In this work, an investigation whose aim is to assess what are the main factors driving the decision on the winner of the Buy Box is performed. For this purpose, a data extraction system is built and 10.906 observations related to twelve different products selected among Italian best-selling categories in the Amazon platform are collected. Data have been acquired during three different time periods: twice a day (noon and late afternoon) for four weeks from February 7 to March 9, 2022; once a day for three weeks from October 20 to November 10, 2022; and once a day for two months (8 weeks) from March 7 to May 6, 2023. The data extraction system represents a scalable solution to extract structured information from Amazon, its code has been released and it is freely available at Github. Subsequently, the extracted data are analyzed by adopting a supervised machine learning approach and by modeling the Buy Box assignment as a classification problem [12]. Precisely, decision trees and rule-based classification algorithms (i.e., *Cart* [14], *Ripper* [48], and *Random Forest* [9]) have been applied in order to compute the most frequent and important features that are present in the classification models. Looking at the results, the features that have been found to mainly influence the winning of the Buy Box are the ratings of the customers, the use of Amazon logistic services (FBA), and being sold by Amazon. Other, non publicly available features (e.g., internal seller rating, stock availability, fees, geographic distance of the product from the customer), may play also an important role.

In conclusion, the paper enters the debate by contributing useful empirical evidence for both research scholars and economists. By means of a better understanding of the main features that drive Amazon's algorithm in choosing the winner of the Buy Box, the functioning of the platform can be assessed and new insights on the topic could be obtained.

The remaining part of the paper is organized as follows. Section 2 introduces the economic background, the Amazon company, its relevance in the e-commerce sector, its leadership, and its main strategies. After briefly reviewing the already-existing literature on the topic, section 3 provides an overview of the developed software system by describing the data extraction process and the empirical investigation. Section 4 presents the performed analyses, presents its main results, and discusses their insights. Finally, section 5 concludes and displays possible future research paths.

2 The prominent role of e-commerce and Amazon

E-commerce has become an established global reality and is an ever-evolving business model. In the past, lack of consumer trust was one of the main barriers to using e-commerce. Today, thanks to changing habits following the Covid-19 pandemic, companies have invested in increasing the online user experience and improving delivery times. According to data collected by [60], 8 out of 10 consumers have maintained the online shopping habits they experienced during the mobility restrictions caused by the pandemic. In this regard, the research analyzes data on the growth of online commerce and highlights the impact of the pandemic on consumers' shopping habits: over a six-month period, most of the consumers who made an online purchase started buying online precisely during the most critical phases of the pandemic emergency. Moreover, this study shows that electronic products and home appliances, clothing, footwear and fashion accessories are the most purchased online. Therefore, an ever-increasing number of people decides to buy online to benefit from the possibility of choosing among a wide range of products at affordable prices, while choosing the payment method and delivery time they prefer and operating within a single platform through quick and easy access. As the number of users increases, so does the supply side which is characterized by many different players: retail operators, marketplaces, B2C and B2B retailers, and brand owners' business partners that provide integrated online services - such as advertising and customer care activities - while ancillary service providers finalize the online shopping experience by managing orders and payments, packaging, and transportation. Online shopping is growing, and as a result, the ecommerce market has spread globally. In a recently released report [34], it is estimated that by 2028, 33% of the world's spending on ecommerce solutions will be cross-border and that this ecommerce model will exceed a global value of $3 trillion within the next five years. In particular, the report says, there will be an increase in purchases, sales and transactions in emerging economies. Marketplaces must grow, and to achieve this they must offer better consumer experiences than the competitor, continually adding value to the service. [52] estimated that 24% of shoppers in 40 countries made cross-border purchases from ecommerce giant Amazon. Alibaba/Aliexpress ranked second with 16% and Shein third with 9%. In addition, according to [44], a phenomenon referred to as the *"grey generation"* of e-commerce is emerging in Asia as 60% more older consumers are active online in Indonesia and 64% more in Thailand.

As explained in [37], e-commerce platforms have two types of participants *"two-sided platforms"*: a business side (B2B), which is the business customer (pays for a service) and an end-user side (B2C), which is the consumer of the service and may pay for it. In this context, each party benefits from the participation of the other party's members. Most platforms focus on sharing common elements among complex products or complex production systems to achieve economies of scale or scope. [28] explains that economies of scale can be achieved for fixed components by increasing production volume, amortizing fixed costs across product types, and using complementary assets (distribution channels or technical support services) more efficiently. [19] cites the example of cases from the past decade (e.g., Airbnb and Uber) to explain how these platforms can develop new capabilities by creating complex systems called *"hybrid and* *multi-sided platforms"* that evolve rapidly over time. However, [58] indicates that the value created by these digital platforms is not only based on their ability to reduce transaction costs and create a meeting place between two parties, but also on four other drivers (trusted environment, data-driven expansion, personalized services and engagement mechanisms). These factors strengthen and enrich the basic value proposition of online e-commerce platforms. In addition, e-commerce makes it possible to strengthen the brand more effectively by building customer loyalty and differentiating from competitors, while gaining more bargaining power when purchasing goods so as to improve pricing strategies and speed up business management processes. According to [50], the factor that has convinced many companies to adopt an e-commerce system is related to the fact that the e-commerce platform allows the same company to be able to integrate its offline business with an online business, maximizing the value of the average purchase. In addition, this new business model can also lead to an increase in offline sales, following the phenomenon called *Webrooming*, where consumers gather information and reviews of the product they intend to purchase on the Internet and then complete the purchase at the physical store. So, e-commerce significantly affects the sales process and consumer habits, speeding up time and improving the shopping experience. Indeed, in this context, errors at the ordering stage are minimized and detailed information about the products or services chosen by the consumer online is updated in real-time.

The spread of the Internet and connected mobile devices has provided people with useful tools to inform themselves online, purchase products and sometimes even express their opinions. Subsequently, the daily use of e-commerce sites has enabled consumers to become more aware of the goods and services for sale, increasing their purchasing power. In this context, therefore, the relationship between the buying consumer and the selling enterprise has changed radically. As explained in [50], the customer over time has taken on the role first of consumer-actor, that is, co-producer of goods together with companies, later in super-consumer with the ability to buy anywhere and judge quality with feedback and reviews, influencing the producing companies. According to [64], digital consumers take on radically different behaviours than the offline generation, avoiding lines in stores and, assuming more power, playing the role of an active party in the relationship with the producing company. So, e-commerce provides an economic advantage both for the consumer in terms of prices, increasing their propensity to buy, and for businesses that have the opportunity to broaden their consumer base.

The purchasing criteria of online consumers can be influenced by many elements. In fact, businesses need to pay attention to consumer behaviour by improving their ability to listen to customer opinions and by better managing the criticism that can be expressed in the form of feedback or reviews. A review is a mode of commentary that allows consumers to tell about their buying experience and the quality of the product after using it, directly on the e-commerce site where they made the purchase. The research [11] found that 95 percent of consumers use reviews and 86 percent consider them an essential part of the purchasing process. Feedback is shorter than a review and is based on the assignment of a number of stars. It is used to express the quality of online transactions. On major e-commerce sites, such as Amazon, the highest rating is five stars, and the lowest is zero. In the past, consumers did not have these tools available to evaluate a shopping experience, so a negative experience directly caused the customer to lose. Today, on the other hand, thanks to this direct exchange of information between consumer and seller, companies have the opportunity to gather important information that they can use to identify business areas in need of improvement, focus on trends that are experiencing more success, and increase online sales volumes. However, there is also another aspect to consider: a negative review or very low feedback for a product or service purchased online could disincentive other consumers from purchasing the same product or service as well, causing a rapid collapse in sales volumes. Therefore, the company must carefully monitor the considerations of their customers and provide a quick response to improve its products and accommodate the needs of the consumers.

The buying process has undergone several transformations since the arrival of the Internet and the first e-commerce sites. Initially, its operation was represented by Procter and Gamble and reported in article [55], through the introduction of the concept of *First Moment Of Truth* (FMOT). According to [43], the term denotes a time frame of a few seconds - from 3 to 7 - in which consumers decide what they want and make a purchase. Later, studies by [38] and [1] were able to identify two new phases related to e-commerce development: the ZMOT or *Zero Moment Of Truth* and *Micro-Moments*. The ZMOT is the moment when an individual picks up his or her laptop, smartphone or any other device and begins to learn about a product or service thinking about trying or buying it. Micro-Moments, on the other hand, represent that brief and critical moment when consumers research online before buying anything and when brands need to be pro-active and establish a direct conversation with their customers. In a society where individuals spend much of their time in front of a smartphone, Micro-Moments become a possible point of contact with a brand and its e-commerce site, influencing the buying process.

As already mentioned before, Amazon Inc. is one of the largest digital players all over the world and especially it is the leading Western company in online commerce. In a nutshell, it is possible to explain this fact as the result of Amazon playing a crucial role in providing access to those digital facilities and infrastructures that nowadays have become essential to run an online business and that are very hard to be replicated (similarly to the rails for a railway system, or the physical infrastructure of the national postal service). As explained in [15], these digital infrastructures of the company or its skill base can be sources of competitive advantage over competitors and these resources cannot always be easily duplicated.

Founded in 1994 by Jeff Bezos, Amazon was originally meant to be an online book store [63]. In just a few years, Amazon had a great development path that allowed it to become the largest web retailer in the world characterized by the most efficient distribution networks in the e-commerce sector. While creating an easy-to-use, safe, and attractive platform for customers all over the world, Amazon has been able to manage thousands of orders every day efficiently and quickly. It has numerous types of products at competitive prices and it almost always tries to keep shipping free. In this way, Amazon builds customer loyalty and captures new consumers on the platform. In addition, through ranking algorithms, Amazon offers a list of preferred products in line with consumer preferences based on the advertisements displayed and the history of previous purchases. The review system on products and sellers allows customers to express their preferences for specific products based on their needs. In this context, highly qualified resources and the ability to foresee future developments in the e-commerce market have allowed Amazon to establish itself globally and achieve the market leader position. Certainly, Amazon's success partially comes from its ability in generating technological innovation within its production processes. Among the others, a leading example is the R&D centre *"Amazon Robotics"* [5] which deals with improving processes through automation systems, robotics, artificial intelligence and the development of recognition systems of objects. Other examples are dynamic pricing systems, robot-driven distribution centres, logistics management of goods and warehouses, and the use of the *"Amazon Lockers"* [4] as a new type of self-service collection points to easy logistic processes. It is worth mentioning that while Amazon's story is usually told as a case of extraordinary entrepreneurial abilities, genius, and merit, we should also take into account many other historical and context-related aspects that have been pivotal for Amazon's growth, e.g., the positive spillovers from the US military R&D sector, the absence of an appropriate regulatory framework for the digital ecosystem, the inability of policy-makers to coordinate at a global level and to catch-up with this fast process of innovation, the role of such a company from a geopolitical viewpoint.

Focusing on its role within e-commerce, Amazon plays a multi-faceted role. It is at the same time the gatekeeper that acts as middleperson in regulating the transactions between buyers and sellers and a seller itself competing with third parties. As a competitor, Amazon can be just a retailer competing with other retailers selling products previously bought by upstream firms, or it can also be a producer by itself. In the latter case the company is at the same time producer, retailer, and owner of the marketplace where it competes with producers of substitute goods. However, there are other examples of vertical integration within the Amazon ecosystem. Indeed, Amazon has developed many complementary services that are necessary inputs for its own business and for one of the other players operating within the platform. The company has entered the market for logistic services *Fulfillment-by-Amazon* (FBA) [3] and the one for pricing algorithms often adopted by third-party sellers. This is just a very short description related to the vertically integrated structure of Amazon Inc. exclusively related to the e-commerce sphere. Indeed, it would be possible to extend the reasoning by looking at its expansion towards many other physical and non-physical markets and industries. Moreover, notice that similar considerations can be done for the other main players within the digital economy. As described in [45] companies such as Uber, Airbnb, Apple or PayPal changed their markets upon entry and today are industry leaders.

Conducting empirical work on the competition for the Amazon Buy Box is therefore reasonable since this is one of the most important lines along which Amazon competes as a seller with third parties. The Amazon Buy Box is the box that appears on the page of a specific product beside description of the product. Through the Buy Box, the majority of consumers conclude their purchase for a specific product within ther Amazon platform. Alternatively, they could decide to buy from one of the other sellers offering the same product by clicking on *other sellers* and going through the list of remaining offers. The third choice would be for customers to leave the platform and buy the product somewhere else. However, given that Amazon is the catalyst of the majority of online commerce and given that choosing a non-winner is more costly than simply clicking one button, then winning the Buy Box plays a crucial role in determining the volume of sales of a given firm and its relevance within the platform.

In order to conduct the empirical investigation, a set of products sold on the Amazon marketplace have been selected. As better explained in Section 4, four different products have been chosen from the best-selling categories: Food, Office, Lighting, and Sports and Leisure. It is worth mentioning that this investigation is specifically related to those products that can be sold on the platform by more than one seller at the same time. Therefore, this study does not concern products whose producer is Amazon itself, i.e., not only products explicitly labelled as Amazon products but also the ones pertaining to brands that have been acquired by Amazon: in this case, Amazon is the unique seller and there is no competition for the Buy Box. The same hold for products uniquely produced and sold by a given company and for the ones sold by companies that have an exclusive dealing contract with Amazon to be the unique seller of that specific product within the marketplace.

3 A platform to predict the winning seller

3.1 Related empirical studies

Before going deeper into the details of the investigation, it is important to have a look at the already existing empirical studies related to the same topic. The empirical literature on these topics is scarce especially if one focuses on the algorithm for the Amazon Buy Box competition. The most closely related works are [29] and [12]. In [29] the authors collect data (price, ratings, assessments, fba, stock availability, recommended choice, time) of several most popular products from the Amazon platform and use a predictive approach based on classification to understand the main drivers of the Buy Box assignment algorithm. They conclude that the ratio between consecutive prices in products and the number of assessments received by customers are the most relevant ones. A similar approach is adopted in [12] where the authors are mainly interested in understanding algorithmic pricing. They derive low prices and feedback of consumers to be the most important factors in determining the winner. Moreover, the authors of [62] explore algorithmic pricing within the platform *Bol.com* and study the effect generated by using algorithmic pricing and the probability of winning the Buy Box. To conclude, there are other empirical investigations dealing with strictly related topics, i.e., evaluating possible issues arising from Amazon's business structure. The authors of [2] study search results on Amazon, trying to understand whether results are diversified across users and which factors drive possible discrimination. In [67], [61], [24], and [39] the authors focus on imitation and the impact on innovation when Amazon enters as competitors. The authors of [13] study self-preferencing by Amazon when recommending joint purchases. In [31] a demand-estimation approach to quantify the welfare effects of both Amazon's dual role and currently proposed regulations is investigated. Even if not published in a peer-reviewed journal, the article reported in [66] is a technical investigation about Amazon's search bar and its search results .

3.2 Data extraction from the web: the Amazon experience

Data extraction from the web consists in an automated procedure that acquires information from non-structured pages of websites, and reorganises them in a structured format [18]. Data extraction from the web can be organised in the following phases [23]

- website analysis: requires examination of the underlying structure of the website, basic understanding of the WWW architecture, Markup languages (e.g. HTML, CSS, XML...) and Web Databases (e.g. MySQL). In this phase, the analyst manually navigates the website and identifies the data of interest that will then have to be acquired automatically, then identifies the paths within the HTML of the page containing this data. XPATH, CSS queries, etc. are used to do this.
- website crawling: involves the development and execution of scripts that automatically explore the site and retrieve data of interest.
- data organisation: storing data into a file (e.g. Excel, CSV, ...) or into a database.

In order to extract data from Amazon, it was chosen to use a technique that precisely manages to extract clean data of interest. An initial inspection of the Amazon web page has been accomplished and two main steps for data acquisition have been performed:

- Crawling: involves the development and execution of algorithms that automatically explore the site and retrieve data of interest;
- Data ingestion: Writing data to a repository as filesystem or a database.

Selenium [27] was used for the crawling step, because it allows the navigation of web pages and it is able to record the steps taken by a human user in order to apply them automatically in future extractions. Selenium contains following components to support browsing automation:

- Selenium IDE, a browser extension for recording and executing browsing without having to write code.
- Selenium Remote Control, a server written in Java that is used to run unit tests on multiple browsers.
- Selenium WebDriver, a driver that allows commands to be executed within the browser, WebDrive can be considered the evolution of Remote Control.

In order to automate page navigation with Selenium, it was necessary to first proceed with the registration of the operations to be performed. The registration of these steps was done with Selenium IDE extension for Chrome or Firefox browsers, which is able to reproduce the same navigation with new snapshot of pages downloaded by the system. These recordings were exported in a format for the Java language and processed by the Java library JSoup [17], an HTML5 parser (compliant with the WHATWG specification) combined with the XPath language [30]. To obtain the data of interest from the web pages, the XPath language was used by defining precise XPath queries. XPath is a language that belongs to the XML family and allows the identification of nodes, i.e. specific parts, within an XML / HTML document. The data extraction pipeline designed in our solution follows the following waterfall steps:

- Start scraping: a batch procedure verifies in the repository of experiments if a new extraction pipeline must be run and produces a message for single page download;
- Scraping: the scraper component receives a message to download a page; it saves the source code of the page in the repository and produces a message for the next step;
- Information extraction: Java with Jsoup extracts data from a raw HTML page and saves them in the repository (the Xpath properties are also collected in the repository).

The steps described above make use of a repository, where to store the acquired data and the extracted information. Indeed, the Google cloud services [6], which allow saving experiments files and information extracted within tabular documents, have been exploited by the use of the Google API. Google Drive and Google Sheets, part of Google Software as a Service (SaaS), were used for persistence. Google Drive is a cloud computing storage and synchronization service. It can be used via the web, uploading and viewing files through the browser, or through an application installed on a computer, which automatically synchronizes a local file system folder with the shared folder, or through Google API. Google Sheet, on the other hand, is an online spreadsheet service that allows information to be processed directly from the browser. Google Sheet was chosen because it allows information to be added to the spreadsheet with the use of API and allows data to be viewed and verified without the need for additional tools but directly from the browser. The Google Drive service was used for saving files in html and png format. The files were stored in folders divided by product. To access these services it was necessary to use a valid Google account. Authentication to the services was performed through the use of the OAuth protocol [65].

3.3 Overview of the system

In this work we use automated data extraction techniques in order to infer the ranking algorithm of Amazon and predict, among the multiple sellers of a product, the winner of the Buy Box. As already mentioned, the Buy Box is the first position in the ranking of sellers and is assigned to the top seller of the rank on the main page of the product. To discover the possible correlations among the ranking position and the information contained in the Amazon webpage, a system for data extraction, data transformation and predictive analysis is needed.

Fundamental for the project was the design and the development of an ad-hoc software to extract data from the amazon.it site. The software was developed using the Java programming language [10]. In addition, three important libraries were used: Selenium [27] with the Google Chrome browser for browsing, Jsoup [17] for parsing the HTML document, and the Google API [6] for uploading data to Google Cloud. Finally, the Google Colab platform [7] was used to carry out the machine learning experiments. All data and software are therefore easily accessible through the Google Drive storage platform.

The system was designed and implemented including state-of-the-art cloud computing components for high-level scalability and adaptability. As it can be seen from Figure

1, modular and reusable technology components have been chosen (e.g. Docker's containers, Kafka, Google IaaS) . Storing data on Google Cloud [6] made these accessible directly through Google's technologies, e.g. Google Colab [7]. We realease the source code of the data extraction system on Github.

Figure 1 depicts the high-level architecture of the adopted solution. The components in the figure show the cloud-based solution, where we wish to leverage the Software as a Services (SaaS) made available by Google Cloud. The acquisition of the information contained in the Amazon pages takes place thanks to XPath-based extraction rules, which are defined a priori [30]. The definition of the rules was performed by an experienced user without the need of unsupervised solutions, which are less expensive, but also less performing. Thanks to the rules, it was always possible to extract the data contained in the pages accurately and store it via the APIs in the cloud repository. Google Cloud services (e.g., Google Compute Engine, Google AppSheet, etc.) were used for the development of the data acquisition pipeline and classification analysis with machine learning (ML) models. The normalised and cleaned data were processed with a feature selection step and then used as training data for different types of ML models, implemented with programming languages (Java, Python, R). To make the solution usable also by less experienced users, a web application was designed by using SaaS tools, where new analyses on Amazon products can be submitted thanks to web interfaces. The solution was designed with robust and scalable microservices [56] and with the use of communication via messaging [36]. It is worth mentioning that, for each component, even the batch code executed server-side for data extraction and for storage, a SaaS service was used in the cloud, ready to be replicated and scaled up as needed. Information-retrival activities were divided into steps by applying the logic of microservices [56]. The services that execute the individual steps were implemented using the Java language with the Spring Boot framework [53] and deployed on docker containers [46]. The execution is orchestrated by means of an event queue, which is managed by Apache Kafka [36].

The UML class diagram of the software is reported in Figure 2. In order to maintain generality, the software does not contain information on the pages to be parsed, but these are read from a document stored in the Google Cloud environment, i.e., a GoogleSheet that is processed through the class *URLRetriever*. After reading the information on the pages to be scanned, the program performs some preliminary operations that consist of performing a navigation on the indicated page in order to display the data of interest. These operations are carried out by the expandPage method of the *Browsing* class. Initially, the driver is connected, which creates an instance of a browser for automatic browsing via the Google Chrome browser, and then the processing of the page is carried out (the elements within the page are identified through paths). The *Experiment* class represents a data acquisition experiment. After *Browsing*, the program downloads the resulting HTML file in the form of a string and parses it with the JSoup library. Since Amazon pages are subject to a high degree of variability, parsing failures are expected, so the software saves the HTML for future processing and for allowing further information to be extracted. Parsing is performed by the methods of the *DataExtractor* class. For proceeding to the data extraction, the file is read with the readXpath method of the *XPathRetriever* class and the data

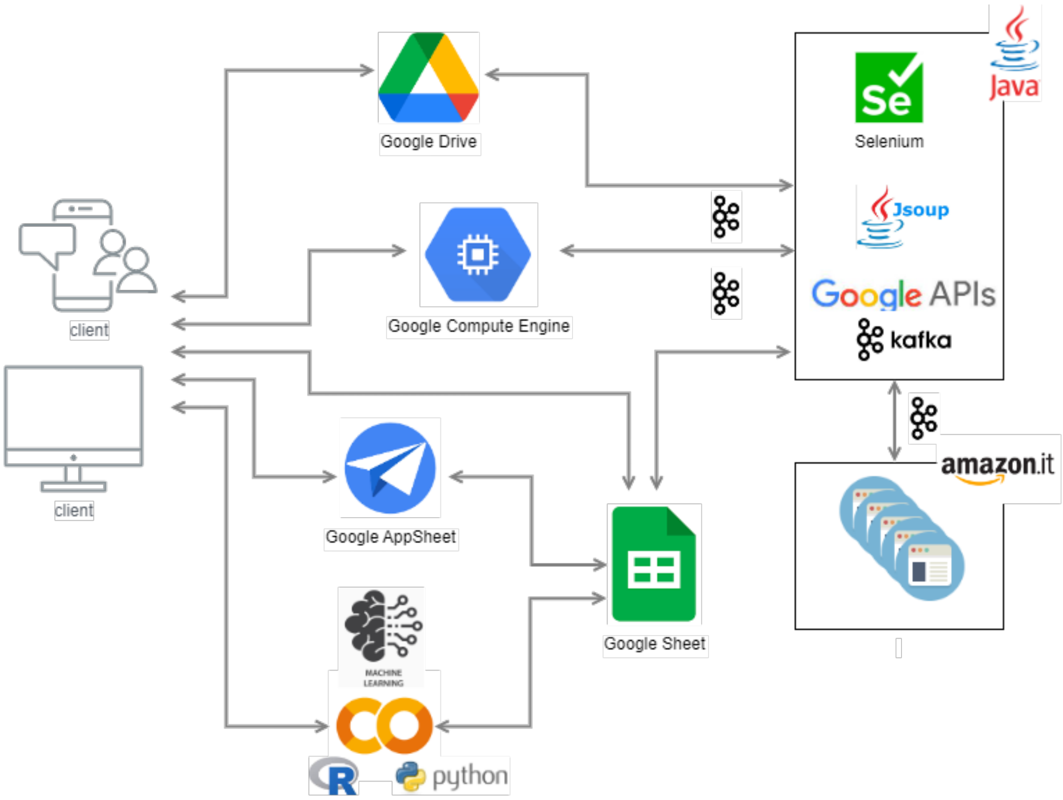

Fig. 1: The high level architecture of the system designed for information retrieval and analysis

is grouped by seller: each vendor is associated with an *OptionProduct* object. At the end of the parsing, structured data is saved on a GoogleSheet with the *id* in the relevant Experiment. To do this, the saveOnGoogleSheet method of the *SheetsOperation* utility class is used. Another utility class is *DriveOperation*, which is used to perform operations on GoogleDrive: file listing, file creation, folder creation and so on. Next, the methods of the *DataExtractor* class are used and the data is grouped by seller: each seller is associated with a *Product* object. A number of additional functionalities have also been implemented: *i*) start scraping on a single product; *ii*) cancel experiment (in case of problems with data extraction); *iii*)load from HTML (allows parsing to be performed on an HTML file, avoiding browsing); *iv*) perform again an experiment (allows re-parsing on a group of files in a date range). *WebScraping* represents the main class with the different start-up options.

4 Results and Discussion

In order to asses the features that play the main role in the ranking algorithm of Amazon marketplace, we focus our data acquisition and analyses on a subset of products

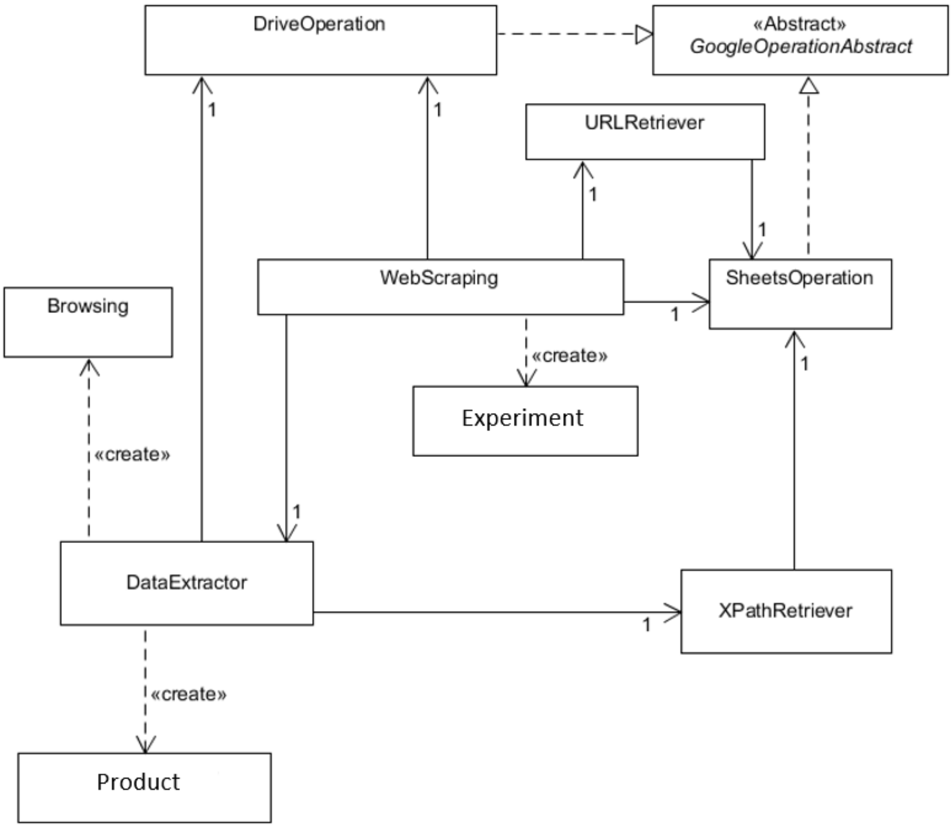

Fig. 2: The UML diagram of the system that describes the classes of the software developed for information retrieval

present in the e-commerce platform. Among four different categories of goods, i.e., food, office, lighting, sports and leisure, we selected the four most sold products in the Italian market and that are offered by at least 8 sellers:

- Food - Borbone Coffee capsules
- Office - Moleskine diary
- Lighting - Philips light bulbs
- Sports and leisure - Smartwatch Xiaomi Mi Smart Band 6

In particular, the Borbone coffee capsules, the Moleskine diary, the Philips light bulbs and the Smartwatch Xiaomi Mi Smart Band 6 were available from 43, 8, 21, and 118 sellers, respectively. These products were also selected because they were considered representative in terms of product category and the interest of buyers. Querying the Amazon platform for the selected products was carried out during two different periods:

- twice a day (at 12:00 noon and 6:00 p.m.) for one month during the time frame between February 7 and March 9, 2022;
- once a day for three weeks from October 20 to November 10, 2022;
- once a day for eight weeks from March 7 to May 6, 2023.

The data collection process was split into microservices, and a total of 10.906 records were collected. We acquired data in form of features and values that can be

buy_box	visibility_order	condition	sold_by	shipped_by	ratings	positive_ratings(%)	price
1	0	New	Amazon	Amazon	40067	0	21
0	1	New	Rck Store	Rck Store	7055	96	20,49
0	2	New	Mondo Copie	Mondo Copie	1603	94	20,99
0	3	New	Ristoservice	Ristoservice	731	86	21
0	4	New	Tipiliano	Tipiliano	40287	96	21,4
0	5	New	ROCARD	ROCARD	15518	99	21,4
0	6	New	JoeBreak2016	JoeBreak2016	5450	99	21,4
0	7	New	3P SHOP	3P SHOP	15371	97	21,5
0	8	New	SAPORE CAFFE	SAPORE CAFFE	769	100	21,5

Fig. 3: A snapshot of the Google Sheet containing the considered data.

found directly on the web page of the selected product. Additionally, we decided to include generated metadata about the acquisition process, and calculated data, i.e., data that can be inferred by combining two or more features and that are calculations about prices, reviews and expected days of delivery. In Figure 3 we depict a snapshot of of the Google Sheet containing the considered data. A detailed list of the features is reported in Table 1. In order to infer the importance of the features in the ranking algorithm of Amazon and consequently for winning the Buy Box, we design the experimentation as a classification problem, where the Buy Box is the dependent feature (class variable) and the other features are the independent ones. The problem may be formulated as the following: given a product offered by n sellers, each of which is characterized by a feature vector, the goal is to classify each seller in winning or not winning the Buy Box, i.e., being assigned to the first position by the ranking algorithm [12]. The classification problem is solved by adopting a supervised learning approach [41]. The learning process is performed by the classification algorithm on a *training set* consisting in labelled samples each one composed of a feature vector (ratings, fba, etc.) and its class (Buy Box winner or not). The inferred classification model (or function) is then applied to a *test set* composed of samples whose class is hidden to the classification algorithm for evaluation purposes or whose class is unknown for prediction purposesco [35]. Finally, the classification task can be defined as the procedure through which an algorithm learns a mapping function (also called model) that assigns a sample to a class [54]. Therefore, the output of a supervised machine learning algorithm is a classification model and the assignment of the samples to a class. State-of-the-art classification algorithms are Linear Regression [51], Naive Bayes [40], Neural Networks [32], K Nearest Neighbour [20], Logistic Regression [33], Perceptrons [49], Relevance Vector Machine (RVM) [57], Support Vector Machines (SVM) [59], Decision Trees [47], Rule-based [22, 8, 14, 16, 25, 26], and Ensembles (Bagging, Boosting, Random forest) [21]. These algorithms can be divided according to the type of classification model they output: a black box model or a human readable one. In this

feature name	*Description*	*origin*
amazon_seller	whether the seller is Amazon	acquired
buy box	indicates whether it is the main selling option or not	generated
condition	whether the product is new or used and, if used, the condition	acquired
d_delivery	days of delivery	acquired
d_shipping	shipping days	acquired
delta_delivery	difference between fastest delivery	calculated
delta_shipping	difference between fastest shipping	calculated
fba	whether the product is shipped by Amazon	acquired
max_d_days	maximum delivery days	calculated
max_e_d_days	maximum expedited delivery days	calculated
min_d_days	minimum delivery days	calculated
min_e_d_days	minimum expedited delivery days	calculated
positive ratings	% of positive ratings in the last 12 months	acquired
price	unit price of the product	acquired
price_diff	price difference from the buy box winner	calculated
price_diff_prod	price difference. (only product)	calculated
price_diff_ship	price difference (only shipping)	calculated
qty_min	minimum quantity of the product sold	acquired
ratings	the number of ratings for the product	acquired
seller rating	the rating assigned to the seller	acquired
shipped by	who is shipping	acquired
shipping price	shipping cost	acquired
shipping type	free or paid	acquired
sold by	the name of the seller	acquired
stars	stars related to the product and seller (from 0 to 5)	acquired
timestamp	timestamp of page snapshot	generated
used condition	the condition of used products	acquired
visibility_order	ranking of seller for the product	acquired

Table 1: Glossary of the features: the considered features in the system.

work, we focus on decision trees and rule-based classification algorithms, whose output is human readable and consists in a classification model composed of *if-then rules* that provide an immediate relationship among the class and the independent features. In this study, we apply three different supervised machine learning algorithms: *Cart* [14], *Ripper* [48], and *Random Forest* [9].
Cart [14] is a decision tree based classification algorithm, where each node, excluding the leaves, represents a value of a feature, based on the value a branch of the tree is chosen and upon reaching the leaf the class is set. The representation of the classification model is a binary tree. Each node represents a single input feature (x) and a split point on that feature (assuming the feature is numeric).The leaf nodes of the tree contain an output variable (y) that corrisponds to the class. The tree is often represented by a graph or a set of rules. New unknown data in input follow the rules of the tree or graph that defines a path to the class label.
Ripper [48] is a rule-based classification algorithm. It derives a set of rules from the training set and produces a model made of if-then rules that can be easily interpreted. Firstly, it tries to derive rules for records that belong to the class C_1. Records belonging to C_1 will be considered as positive examples($+ve$) and other classes will be considered as negative examples($-ve$). The sequential Covering Algorithm is used to generate the rules that discriminate between $+ve$ and $-ve$ examples. In the following steps Ripper tries to derive rules for the other class C_2. The process is repeated until stopping criteria is met.
Random Forest [9] is an ensemble learning method for classification that creates a multitude of single and different decision trees at training time. The algorithm is able

to generate many decision trees by using a subsets of the input and the final classification is performed by combining all generated trees and by computing the classification according to the class selected by most trees. Typically, a hundred to several thousand trees are used, depending on the size and on the nature of the training set. An optimal number of trees can be found by using valid sampling techniques. For this reason, Random Forests generally outperform single decision trees.

For performing the machine learning analysis and knowledge extraction a data processing procedure was implemented, where new product-independent features were created, i.e., calculated features reported in Table 1.

In order to remove redundancy, we selected a set of representative features related to each specific category, i.e. price, ratings, shipping, seller. Additionally, the sale price was transformed into new calculated features price_diff, price_diff_prod, price_diff_ship: the difference of the price of the considered seller with the price of the same product that is offered by the seller that wins the Buy Box (total price, only the product price, and only the shipping price). We excluded specific information related to the product, such as the price, by using more objective features, such as price differences.
In particular, following feature selection steps were applied: i) to exclude features with a high number of missing values (i.e., greater than 60%) in order to include only informative features with a low number of missing values; the excluded features were used_condition and seller rating. ii) to remove information redundancy a set of representative features related to each specific category (i.e., price, ratings, shipping, seller) was chosen; the excluded feature were: delta_delivery, the d_delivery (days of delivery) feature was used; delta_shipping, the days of shipping feature was used; price_unit; shipping price, the price_diff_ship feature was used; max_d_days, max_e_d_days, min_d_days, min_e_d_days were used for calculating the delta delivery, i.e., the difference between the current and the fastest delivery; iii) to remove non informative feature that do not contribute to the classification process; the excluded features were condition, positive ratings (% of positive ratings in the last 12 months) - conversely the stars were chosen, rating of the seller (seller ratings are not available when Amazon itself acts as a seller on its platform), shipped by, shipping type (free or paid - acquired), and sold by (the name of the seller). It is worth noting that timestamp and visibility order were used to get track of the data acquisition and the order of the sellers, respectively. These two variables were used as metadata for the classification experiments. The set of features used for the analysis are reported in Table 2.

In order to obtain a balanced dataset, it was chosen not to take all records into account, which would have led to an imbalanced dataset, because of the higher presence of non Buy Box winning sellers. For this reason, it was chosen to take the record of the Buy Box winner and the next n records of non-winners in order of ranking. The ratio between the Buy Box vs non Buy Box classes was set to 1:2. The choice of non Buy Box records was performed randomly from a sample of the top-5 ordered by ranking, i.e., the next five records below the Buy Box winner were considered, and from these, a random choice method identified two records in each run. Moreover, a normalization algorithm was applied to the collected data to convert the values of numerical features in a 0:1 scale.

feature name	*min*	*max*	*avg*	*st.dev*
buy_box	no	yes	-	-
d_delivery	0 (-5)	1 (20)	0,3 (2,57)	0,16 (3.94)
d_shipping 0 (0)	1 (13)	0,12 (1,58)	0,19 (2,43)	
fba	no	yes	-	-
price_diff	0 (0)	1 (40,48)	0,18 (7,26)	0,17 (6,8)
price_diff_prod	0 (0)	1 (20,81)	0,33 (6,96)	0,25 (5,27)
price_diff_ship	0 (0)	1 (13,40)	0,08 (1,09)	0,18 (2,37)
qty_min	1 (1)	3 (3)	1,01	0,15
ratings	0 (0)	1 (112.466)	0,11 (12.862)	0,17 (19442,57)
amazon_seller	no	yes	-	-
stars	0 (0)	1 (5)	0,85 (4,23)	0,25 (1,24)

Table 2: The 11 features used for the machine learning analysis and their ranges of normalized (and raw) values.

For each supervised classification algorithm and for each observed period three runs with different random seeds and with different sampling ratios of training and testing (i.e. 60%-40%, 70%-30%, 80%-20%, 100% full training set) were performed, obtaining 12 runs for each classification algorithm and for each observed period, resulting in 108 classification experiments. We release the source code of the analysis framework on a public repository of GitHub.

In Table 3 we report the most relevant features obtained from the supervised machine learning analysis by highlighting the features that are present in the classification models ranked in order of importance. Thanks to the *variable importance* (*varImp*) method of the used R library, we report the ranking of the features in order of importance for each classification algorithm. This method is used in Machine Learning to rank the importance of the features in a classification problem and model [9]. The method performs a statistical test based on the occurrences and the classification performance of each specific feature present in the model. It generates an importance index, more specifically a floating-point number, that determines whether the feature affects more (higher value) or less (lower value) the Buy Box assignment. For each analysis period and for each ML model (e.g. CART, Random Forest, and Ripper), all importance results were pooled by calculating the overall average percentage for each feature. This percentage calculation on individual features by model and period generated a top-10 ranking results shown in Table 3. Looking at the data in Table 3, it is worth noting that the features ratings, amazon_seller, and fba are more important over the other features, when considering the Buy Box assignment as a classification problem.

CART and *Random Forest*, both based on decision trees, place ratings, amazon_seller, and fba at the top while *Ripper* brings the ratings and price_diff features to the top. Surprisingly, price is not always at the top as we might have thought. This may mean a possible strong influence of other factors, such as *ratings* and *fba* (shipped by Amazon), in the allocation of the Buy Box.

Figure 4 summarizes the findings and shows the feature importance results weighted as an aggregation of all calculated importance percentages. Again, it can be noticed that ratings, fba, and sold by Amazon are the most relevant features for the

Rank	*CART*	*Random* Forest	*Ripper*	*Final*
		Experiment 1		
1	ratings (26%)	ratings (37%)	ratings (61%)	**ratings (41%)**
2	amazon_seller (23%)	amazon_seller (26%)	price_df_prod (15%)	**amazon_seller (18%)**
3	fba (18%)	fba (14%)	d_delivery (11%)	**fba (12%)**
4	d_shipping (13%)	price_diff (6%)	stars (6%)	**d_shipping (6%)**
5	price_diff (11%)	d_shipping (5%)	price_df_ship (4%)	**price_diff (6%)**
6	price_df_prod (5%)	price_df_prod (5%)	fba (3%)	**d_delivery (5%)**
7	stars (2%)	stars (4%)	qty_min (0%)	**price_df_prod (5%)**
8	d_delivery (2%)	d_delivery (3%)	price_diff (0%)	**stars (3%)**
9	price_df_ship (1%)	price_df_ship (0%)	sold_amazon (0%)	**price_df_ship (0%)**
10	qty_min (0%)	qty_min (0%)	d_shipping (0%)	**qty_min (0%)**
		Experiment 2		
1	ratings (30%)	ratings (39%)	ratings (65%)	**ratings (44%)**
2	amazon_seller (29%)	amazon_seller (30%)	fba (23%)	**amazon_seller (21%)**
3	fba (21%)	fba (14%)	price_df_prod (6%)	**fba (19%)**
4	d_shipping (10%)	price_df_prod (6%)	price_diff (3%)	**price_df_prod (7%)**
5	price_df_prod (9%)	d_shipping (4%)	sold_amazon (3%)	**d_shipping (5%)**
6	d_delivery (0%)	price_diff (4%)	stars (0%)	**price_diff (2%)**
7	price_df (0%)	d_delivery (2%)	price_df_ship (0%)	**d_delivery (1%)**
8	price_df_ship (0%)	stars (1%)	d_delivery (0%)	**stars (1%)**
9	stars (0%)	price_df_ship (0%)	d_shipping (0%)	**price_df_ship (0%)**
10	qty_min (0%)	qty_min (0%)	qty_min (0%)	**qty_min (0%)**
		Experiment 3		
1	amazon_seller (58%)	amazon_seller (50%)	amazon_seller (47%)	**amazon_seller (52%)**
2	ratings (22%)	ratings (28%)	ratings (11%)	**ratings (20%)**
3	d_shipping (8%)	d_shipping (8%)	price_diff_prod (10%)	**d_shipping (8%)**
4	price_diff_prod (7%)	price_df_prod (7%)	d_shipping (8%)	**price_df_prod (8%)**
5	stars (4%)	price_diff (3%)	price_diff (7%)	**price_diff (4%)**
6	price_diff (1%)	stars (1%)	stars (7%)	**stars (4%)**
7	fba (0%)	price_df_ship (1%)	fba (5%)	**price_df_ship (2%)**
8	price_df_ship (0%)	fba (0%)	price_df_ship (0%)	**fba (2%)**
9	d_delivery (0%)	d_delivery (0%)	d_delivery (0%)	**d_delivery (0%)**
10	qty_min (0%)	qty_min (0%)	qty_min (0%)	**qty_min (0%)**

Table 3: Importance of the features extracted from the machine learning models in the different experiments.

Buy Box assignment process. Additionally, although the data are acquired during different periods, the result are similar. In essence, the results obtained in the first period are confirmed in the second and third period as well.

Finally, the performance of the classification models was also evaluated in every performed run of the analyses: the accuracy of each generated model was measured and reached always a value that was greater than 96%.

5 Conclusions

In this work, we presented a modern and scalable solution for extracting clean and structured information from the e-commerce platform of Amazon. We designed and developed also an analysis framework based on supervised machine learning algorithms in order to infer which features influence at most the assignment of the Buy Box, i.e., the default shopping option in the Amazon platform. Analyses were conducted on a set of selected products among the most sold in Italy and data were collected during three different periods. After the analysis with three different supervised machine learning algorithms and several experimental runs, similar results were obtained in different observation periods. By measuring the importance of the features influencing the Buy

Box assignment in this experimentation, we obtained interesting results: number of ratings, sold by Amazon, and fba (i.e. shipped by Amazon) are the most important features according to our analyses. The results show that those features play a very important role in the Buy Box assignment process of the Amazon platform.

Regarding the rating, it is not surprising that this feature plays the most important role in the Buy Box assignment process according to our analysis. Indeed, to increase sales on Amazon, it is necessary for the seller to provide perfect products and customer service that the user can evaluate through two different types of feedback: the rating of the product and the rating of the seller. As explained by [42], the so-called *"social proof"* has a strong influence on the customer's final purchase decision. In addition, through a detailed review a rating can provide the customer with additional information that is not contained in the description, because it is the result of the actual use of the product. According to the survey performed by the authors of [11], an accurate description and good reviews and ratings will help customers make better purchase decisions and, as a result, have a positive impact on sales. A rating ranging from 1 star to 5 stars is added to the products according to users' reviews. Thus, a good quantity and quality of Amazon reviews for a given product has a crucial impact on revenue. For this reason, the most common sales strategy on Amazon is to encourage and to collect customer reviews. Also because product filtering is an integral part of the buying process on the Amazon platform and one of the most widely used filters is *"average customer ratings"*. The very high number of reliable reviews left on products, in fact, makes purchases much safer, as it reduces the risk of buying a product that does not correspond to its description. For this reason, the Amazon platform is considered very reliable. Therefore, a particular product without positive reviews or ratings will gain little visibility. Ratings of the seller, on the other hand, related to the individual seller's *"Amazon Seller"* account, but are not tied to a particular product. After purchasing a product, in fact, the customer has the opportunity to share feedback about the seller, evaluating its performance at all stages of the buying experience. Parameters such as the speed of shipping if the item arrived on time and matched the description, as well as the timeliness of customer service are evaluated. Also in the case of the rating of the seller, a range from 1 to 5 stars can be given. Ratings to the seller appear when a user clicks on *"sold by"* on a product's page: the user is redirected, to a page where, in addition to information about the seller, he will find the quantity, the average score of feedback received and the reviews posted in the last year. Product and seller reviews are equally important, although product feedback has a greater influence at the sales level because users consult it much more frequently than seller feedback. This is because of the product-centric nature of Amazon. Generally, it is rare for a user to read the reviews of a seller, unless the user is interested in a very expensive product that requires a thorough evaluation of the purchase. In most cases, the user navigates on the product page and if the reviews are good and the score is high, he proceeds to purchase. On the other hand, seller feedback is critical for Amazon seller account management: the presence of negative seller ratings, which negatively affect revenue, could also lead Amazon to temporarily suspend the account or, as an extreme consequence, prevent the sale of other products altogether. In this regard, it is worth noting that seller ratings are not available when Amazon itself acts as a seller on its platform. To conclude, the results of our analysis showed also that Amazon seller (i.e., sold by Amazon) and fba (i.e., shipped by Amazon) are features that are relevant for the Buy Box assignment in the Amazon platform.

Finally, we wish to highlight that, there are other features possibly used by the algorithm of Amazon to assign the Buy Box, e.g., sales volume, response time to customer inquiries, rate of returns and refunds, and previous shipping times. These features are not publicly available and therefore not considered in this work. It would be interesting in future studies to acquire such features and include them in the analysis, if possible. Hence, some strategic features influencing the decision of the algorithm may require further analyses and investigations.

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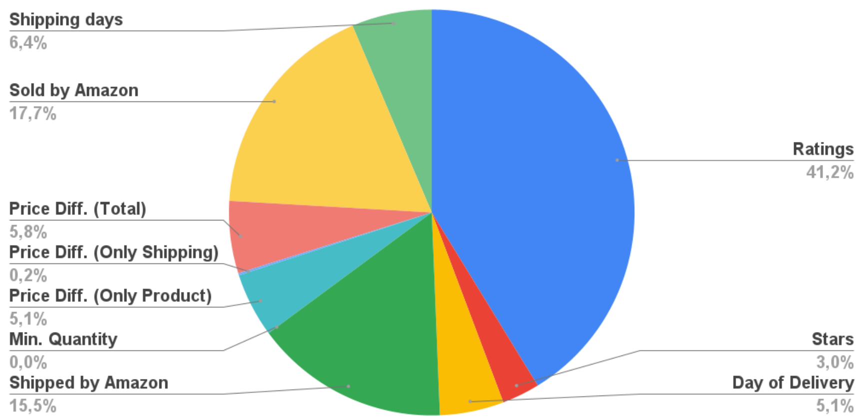
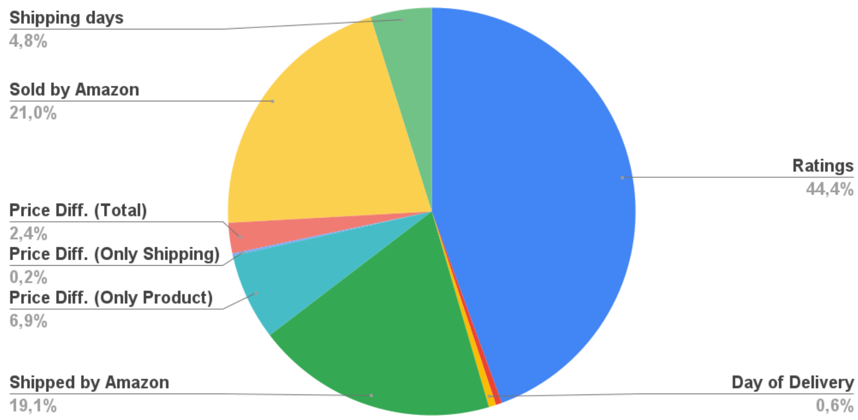
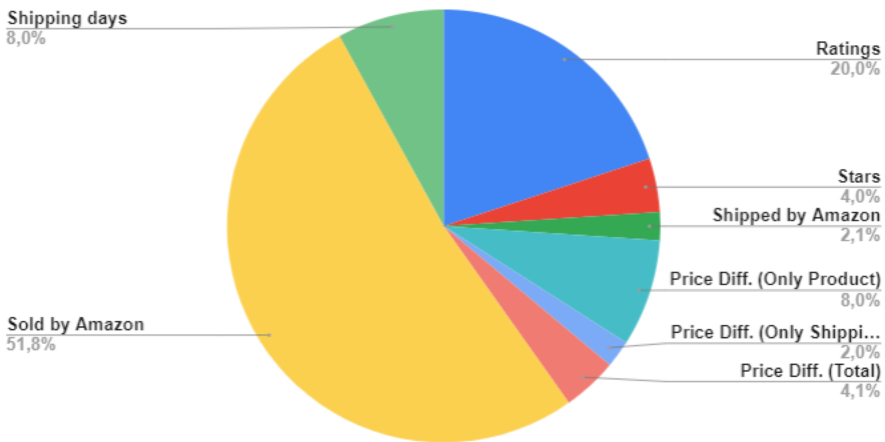


(a) Most important features in Exp. 1



(b) Most important features in Exp. 2



(c) Most important features in Exp. 3

Fig. 4: Feature importance: final results